

Class 18: Pertussis and the CMI-PB project

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Investigating pertussis cases by year

```
library(datapasta)
library(tibble)
library(ggplot2)
```

```
cdc <- tribble(
  ~Year, ~Number.of.Reported.Pertussis.Cases, ~Data.Status,
  1922L, 107473, "Finalized",
  1923L, 164191, "Finalized",
  1924L, 165418, "Finalized",
  1925L, 152003, "Finalized",
  1926L, 202210, "Finalized",
  1927L, 181411, "Finalized",
  1928L, 161799, "Finalized",
  1929L, 197371, "Finalized",
  1930L, 166914, "Finalized",
  1931L, 172559, "Finalized",
  1932L, 215343, "Finalized",
  1933L, 179135, "Finalized",
  1934L, 265269, "Finalized",
```

1935L,	180518,	"Finalized",
1936L,	147237,	"Finalized",
1937L,	214652,	"Finalized",
1938L,	227319,	"Finalized",
1939L,	103188,	"Finalized",
1940L,	183866,	"Finalized",
1941L,	222202,	"Finalized",
1942L,	191383,	"Finalized",
1943L,	191890,	"Finalized",
1944L,	109873,	"Finalized",
1945L,	133792,	"Finalized",
1946L,	109860,	"Finalized",
1947L,	156517,	"Finalized",
1948L,	74715,	"Finalized",
1949L,	69479,	"Finalized",
1950L,	120718,	"Finalized",
1951L,	68687,	"Finalized",
1952L,	45030,	"Finalized",
1953L,	37129,	"Finalized",
1954L,	60886,	"Finalized",
1955L,	62786,	"Finalized",
1956L,	31732,	"Finalized",
1957L,	28295,	"Finalized",
1958L,	32148,	"Finalized",
1959L,	40005,	"Finalized",
1960L,	14809,	"Finalized",
1961L,	11468,	"Finalized",
1962L,	17749,	"Finalized",
1963L,	17135,	"Finalized",
1964L,	13005,	"Finalized",
1965L,	6799,	"Finalized",
1966L,	7717,	"Finalized",
1967L,	9718,	"Finalized",
1968L,	4810,	"Finalized",
1969L,	3285,	"Finalized",
1970L,	4249,	"Finalized",
1971L,	3036,	"Finalized",
1972L,	3287,	"Finalized",
1973L,	1759,	"Finalized",
1974L,	2402,	"Finalized",
1975L,	1738,	"Finalized",
1976L,	1010,	"Finalized",

1977L,	2177,	"Finalized",
1978L,	2063,	"Finalized",
1979L,	1623,	"Finalized",
1980L,	1730,	"Finalized",
1981L,	1248,	"Finalized",
1982L,	1895,	"Finalized",
1983L,	2463,	"Finalized",
1984L,	2276,	"Finalized",
1985L,	3589,	"Finalized",
1986L,	4195,	"Finalized",
1987L,	2823,	"Finalized",
1988L,	3450,	"Finalized",
1989L,	4157,	"Finalized",
1990L,	4570,	"Finalized",
1991L,	2719,	"Finalized",
1992L,	4083,	"Finalized",
1993L,	6586,	"Finalized",
1994L,	4617,	"Finalized",
1995L,	5137,	"Finalized",
1996L,	7796,	"Finalized",
1997L,	6564,	"Finalized",
1998L,	7405,	"Finalized",
1999L,	7298,	"Finalized",
2000L,	7867,	"Finalized",
2001L,	7580,	"Finalized",
2002L,	9771,	"Finalized",
2003L,	11647,	"Finalized",
2004L,	25827,	"Finalized",
2005L,	25616,	"Finalized",
2006L,	15632,	"Finalized",
2007L,	10454,	"Finalized",
2008L,	13278,	"Finalized",
2009L,	16858,	"Finalized",
2010L,	27550,	"Finalized",
2011L,	18719,	"Finalized",
2012L,	48277,	"Finalized",
2013L,	28639,	"Finalized",
2014L,	32971,	"Finalized",
2015L,	20762,	"Finalized",
2016L,	17972,	"Finalized",
2017L,	18975,	"Finalized",
2018L,	15609,	"Finalized",

```

2019L,      18617, "Finalized",
2020L,      6124, "Finalized",
2021L,      2116, "Finalized",
2022L,      3044, "Finalized",
2023L,      7063, "Finalized",
2024L,     43321, "Provisional",
2025L,     28783, "Provisional"
)

```

```
colnames(cdc)
```

```

[1] "Year"                                "Number.of.Reported.Pertussis.Cases"
[3] "Data.Status"

```

```
head(cdc)
```

```

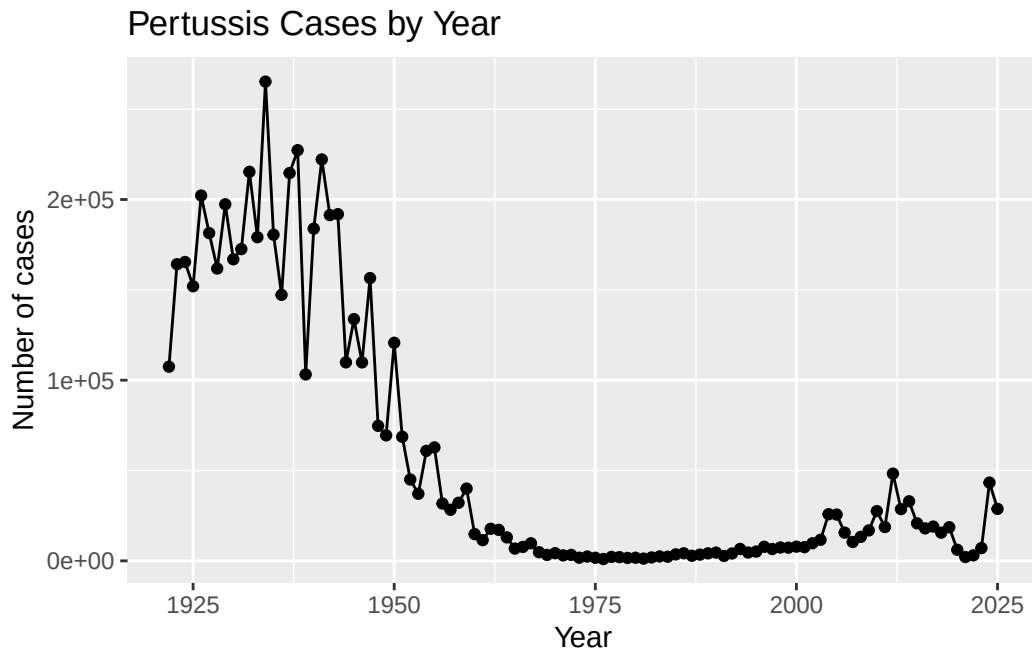
# A tibble: 6 x 3
  Year Number.of.Reported.Pertussis.Cases Data.Status
<int>                                <dbl> <chr>
1  1922                                107473 Finalized
2  1923                                164191 Finalized
3  1924                                165418 Finalized
4  1925                                152003 Finalized
5  1926                                202210 Finalized
6  1927                                181411 Finalized

```

```

ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  labs(
    title = "Pertussis Cases by Year",
    x = "Year",
    y = "Number of cases"
  )

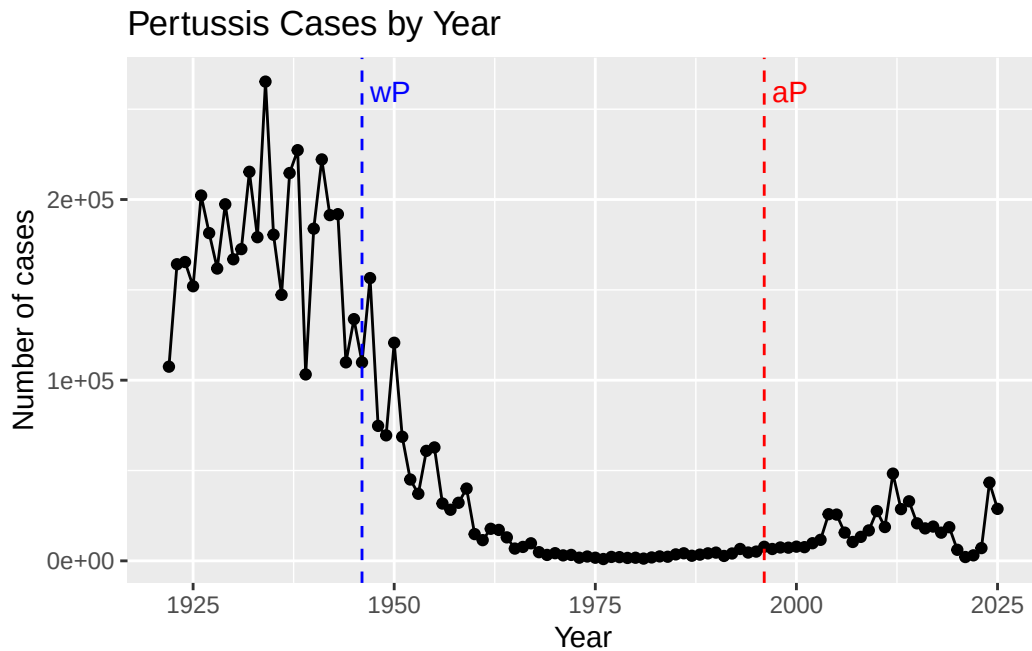
```



A tale of two vaccines (wP & aP)

Q2. Using the ggplot geom_vline() function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, linetype = "dashed", color = "blue") +
  geom_vline(xintercept = 1996, linetype = "dashed", color = "red") +
  annotate("text", x = 1947, y = 260000, label = "wP", color = "blue", hjust = 0) +
  annotate("text", x = 1997, y = 260000, label = "aP", color = "red", hjust = 0) +
  labs(
    title = "Pertussis Cases by Year",
    x = "Year",
    y = "Number of cases"
  )
```



Pertussis cases dropped sharply after the introduction of wP vaccine and slightly risen again after the introduction of aP

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine, there was slight increase in pertussis cases compared to wP vaccine, while this still wasn't significantly high. The possible explanation is specificity of aP vaccine and shorter lasting protection

Exploring CMI-PB data

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)

head(subject, 3)
```

subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not Hispanic or Latino	White

2	2	wP	Female Not Hispanic or Latino White
3	3	wP	Female Unknown White
	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female  Male
112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$biological_sex, subject$race)
```

	American Indian/Alaska Native	Asian	Black or African American
Female	0	32	2
Male	1	12	3

	More Than One Race	Native Hawaiian or Other Pacific Islander
Female	15	1
Male	4	1

	Unknown or Not Reported	White
Female	14	48
Male	7	32

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
today()
```

```
[1] "2026-06-08"
```

```
today() - ymd("2000-01-01")
```

Time difference of 9655 days

```
time_length( today() - ymd("2000-01-01"), "years")
```

```
[1] 26.43395
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
library(lubridate)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union


```
subject$age <- today() - ymd(subject$year_of_birth)
```

```
ap <- subject %>% filter(infancy_vac == "aP")
round( summary( time_length( ap$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
wp <- subject %>% filter(infancy_vac == "wP")
round( summary( time_length( wp$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

```
t.test(time_length(age, "years") ~ infancy_vac, data = subject)
```

Welch Two Sample t-test

data: time_length(age, "years") by infancy_vac

t = -12.918, df = 104.03, p-value < 2.2e-16

alternative hypothesis: true difference in means between group aP and group wP is not equal to 0

95 percent confidence interval:

-10.094058 -7.407351

sample estimates:

mean in group aP mean in group wP

28.3293 37.0800

Q8. Determine the age of all individuals at time of boost?

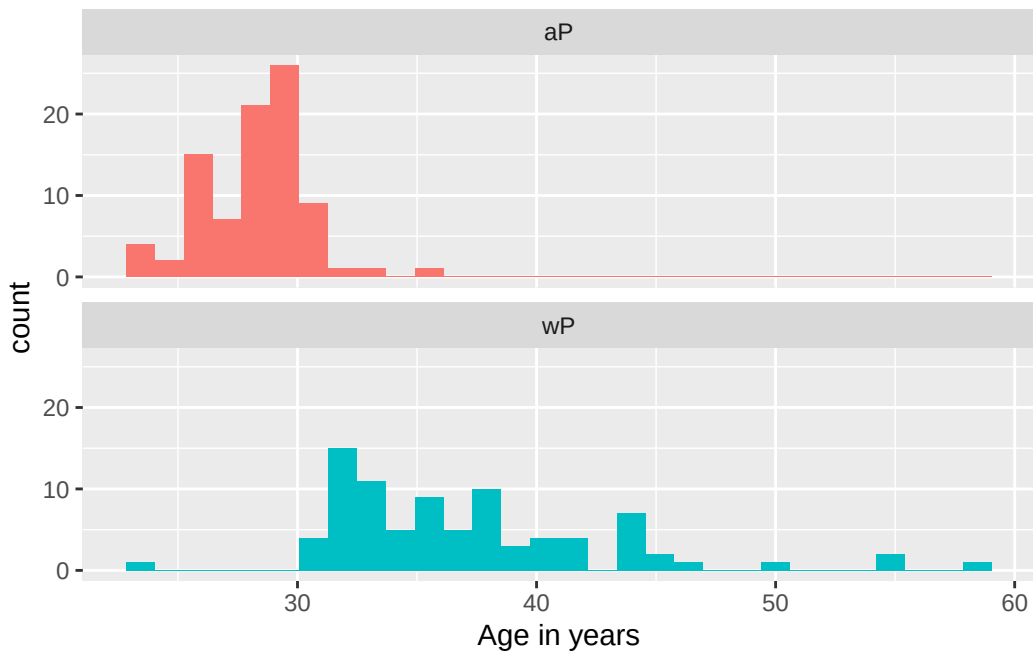
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
x <- t.test(time_length( wp$age, "years" ),
            time_length( ap$age, "years" ))

x$p.value
```

```
[1] 2.372101e-23
```

```
specimen <- read_json(
  "https://www.cmi-pb.org/api/specimen",
  simplifyVector = TRUE
)
```

```
titer <- read_json(
  "https://www.cmi-pb.org/api/plasma_ab_titer",
  simplifyVector = TRUE
)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(specimen, subject, by = "subject_id")

dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

```
specimen_id subject_id actual_day_relative_to_boost
1           1           1                        -3
2           2           1                         1
3           3           1                         3
4           4           1                         7
5           5           1                        11
6           6           1                        32
planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood      1          wP         Female
2                             1         Blood      2          wP         Female
3                             3         Blood      3          wP         Female
4                             7         Blood      4          wP         Female
5                            14         Blood      5          wP         Female
6                            30         Blood      6          wP         Female
ethnicity race year_of_birth date_of_boost dataset
1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
age
1 14768 days
2 14768 days
```

```
3 14768 days
4 14768 days
5 14768 days
6 14768 days
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta, by = "specimen_id")

dim(abdata)
```

```
[1] 52576    21
```

```
head(abdata)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	UG/ML	2.096133	1	-3
2	IU/ML	29.170000	1	-3
3	IU/ML	0.530000	1	-3
4	IU/ML	6.205949	1	-3
5	IU/ML	4.679535	1	-3
6	IU/ML	2.816431	1	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

```

5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
  age
1 14768 days
2 14768 days
3 14768 days
4 14768 days
5 14768 days
6 14768 days

```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```

IgE  IgG  IgG1  IgG2  IgG3  IgG4
6698 5389 10117 10124 10124 10124

```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```

2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520           8085           7301           5670

```

The most recent dataset has the fewest rows (newer dataset may have fewer available antibody titer measurements)

Examine IgG Ab titer levels

```

igg <- abdata %>% filter(isotype == "IgG")
head(igg)

```

```

specimen_id isotype is_antigen_specific antigen      MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614         3.736992
2           1     IgG                TRUE     PRN  332.12718         2.602350
3           1     IgG                TRUE     FHA 1887.12263        34.050956
4          19     IgG                TRUE      PT   20.11607         1.096366

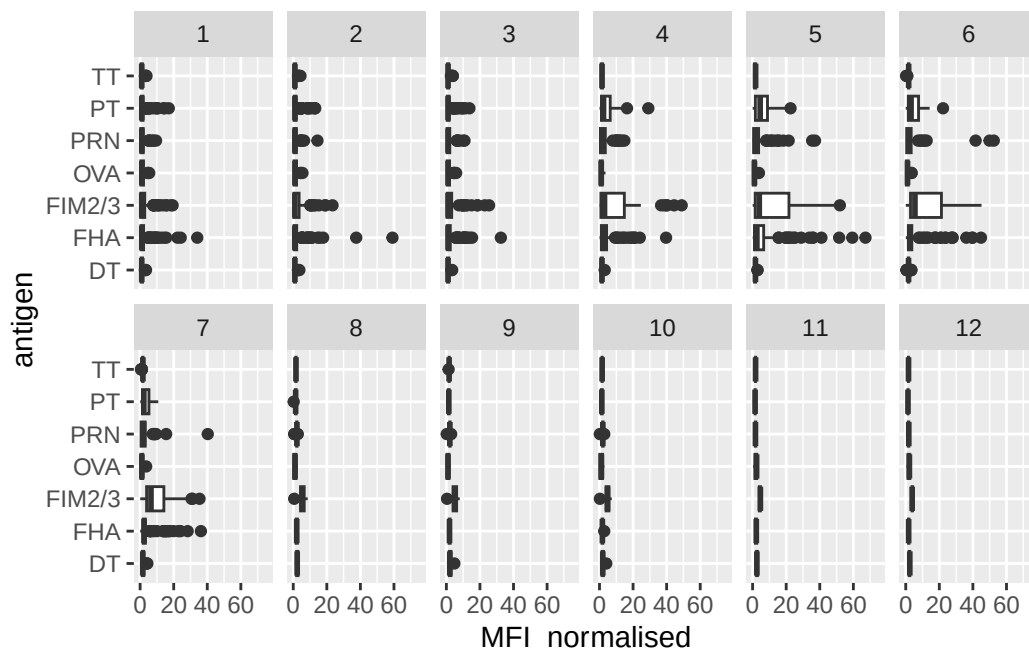
```

5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457
	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost		
1	IU/ML	0.530000	1			-3
2	IU/ML	6.205949	1			-3
3	IU/ML	4.679535	1			-3
4	IU/ML	0.530000	3			-3
5	IU/ML	6.205949	3			-3
6	IU/ML	4.679535	3			-3
	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex	
1	0	Blood	1	wP	Female	
2	0	Blood	1	wP	Female	
3	0	Blood	1	wP	Female	
4	0	Blood	1	wP	Female	
5	0	Blood	1	wP	Female	
6	0	Blood	1	wP	Female	
	ethnicity	race	year_of_birth	date_of_boost	dataset	
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
	age					
1	14768 days					
2	14768 days					
3	14768 days					
4	15864 days					
5	15864 days					
6	15864 days					

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(x = MFI_normalised, y = antigen) +
  geom_boxplot() +
  xlim(0, 75) +
  facet_wrap(vars(visit), nrow = 2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

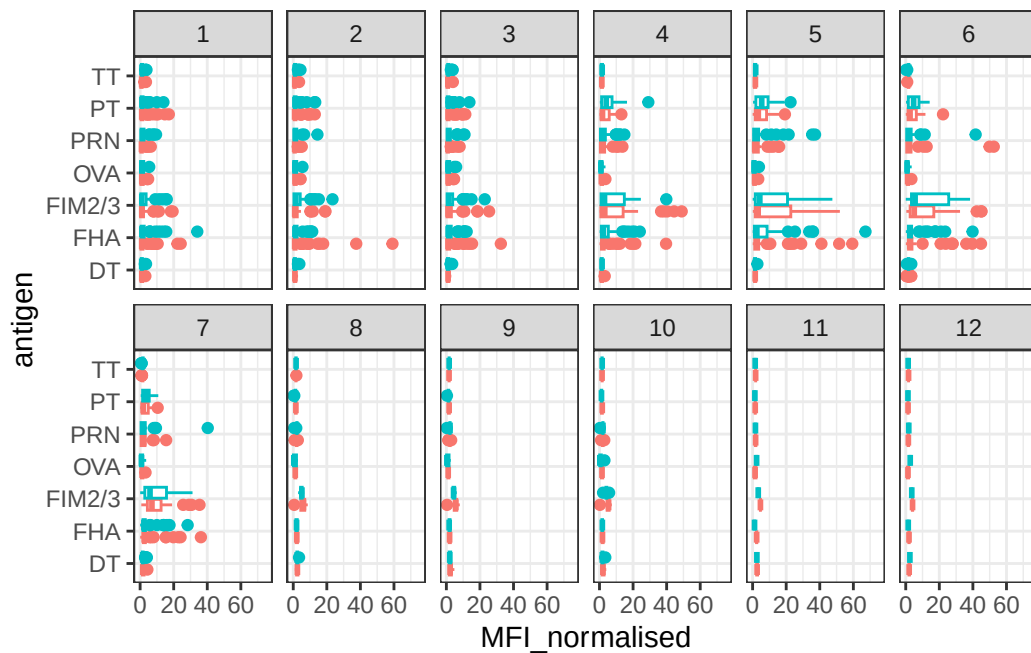


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

PT, FIM2/3, and FHA show clear differences in the level of IgG antibody titers recognizing them over time. These likely change because they are pertussis-related vaccine antigen. Others, especially OVA, do not change because they are not pertussis vaccine.

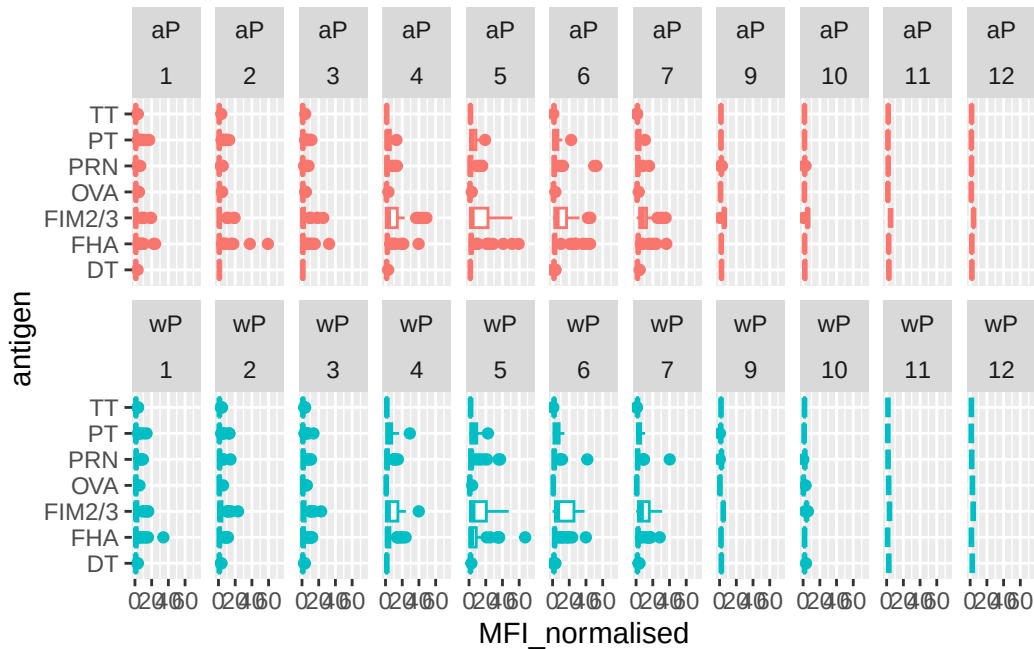
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (``stat_boxplot()``).



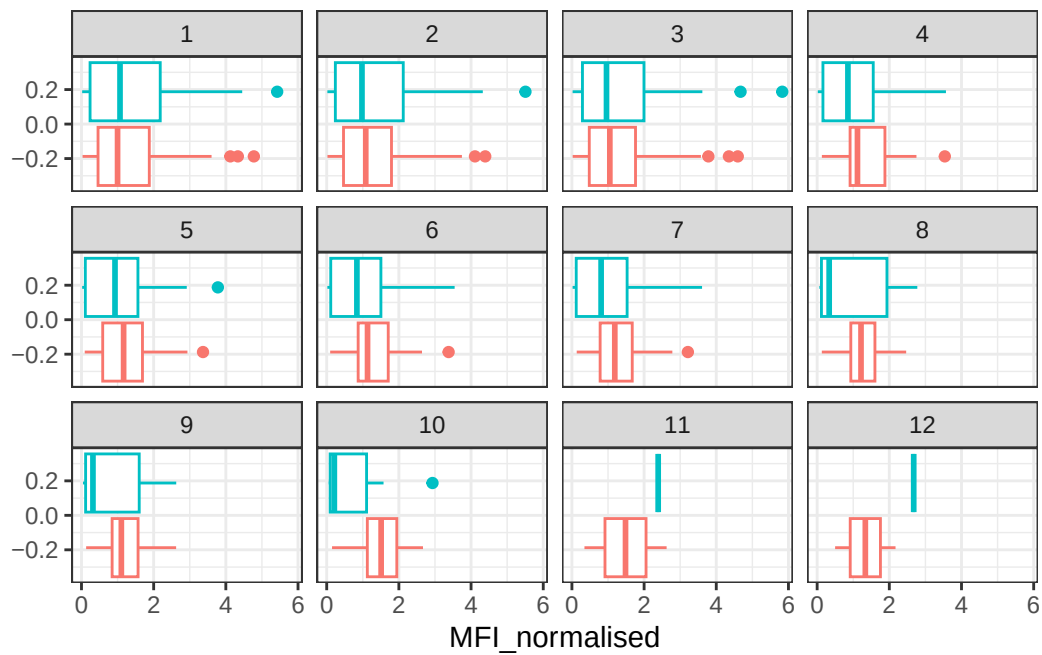
```
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

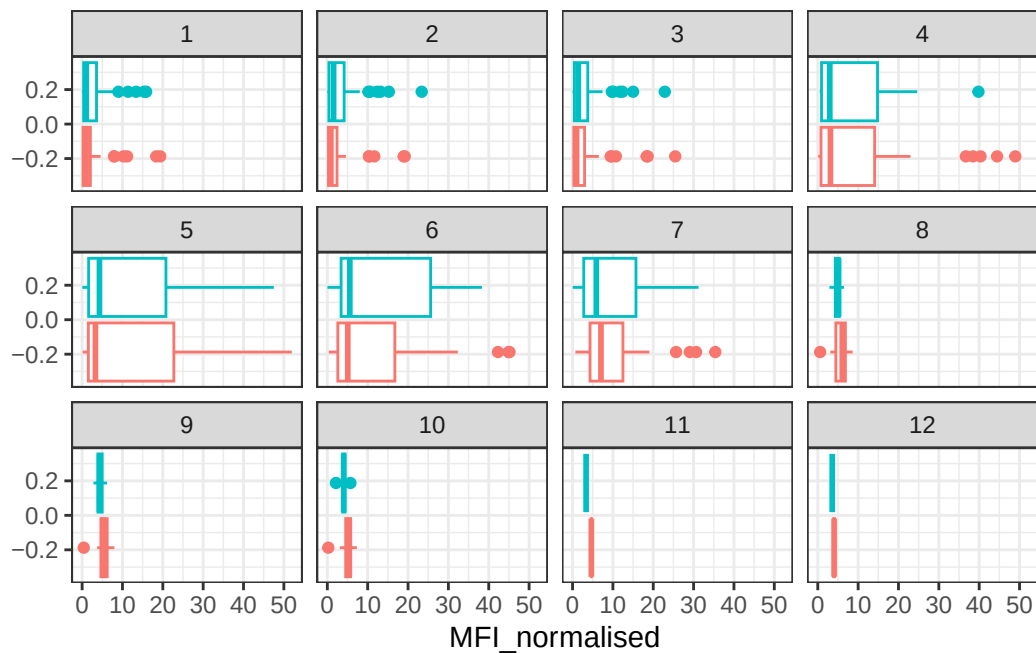


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

```
filter(igg, antigen == "OVA") %>%
  ggplot() +
  aes(x = MFI_normalised, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen == "FIM2/3") %>%
  ggplot() +
  aes(x = MFI_normalised, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```

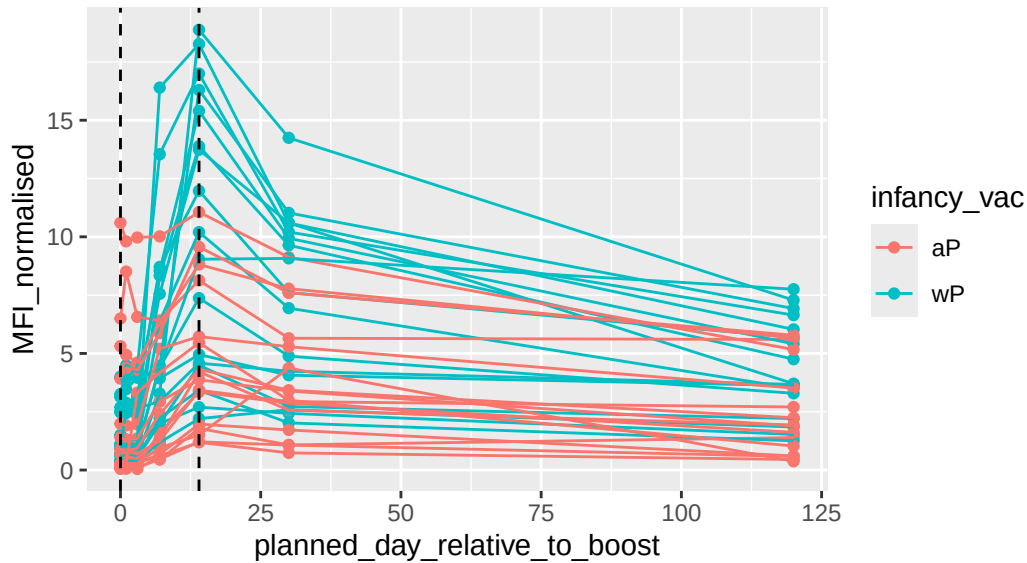


```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



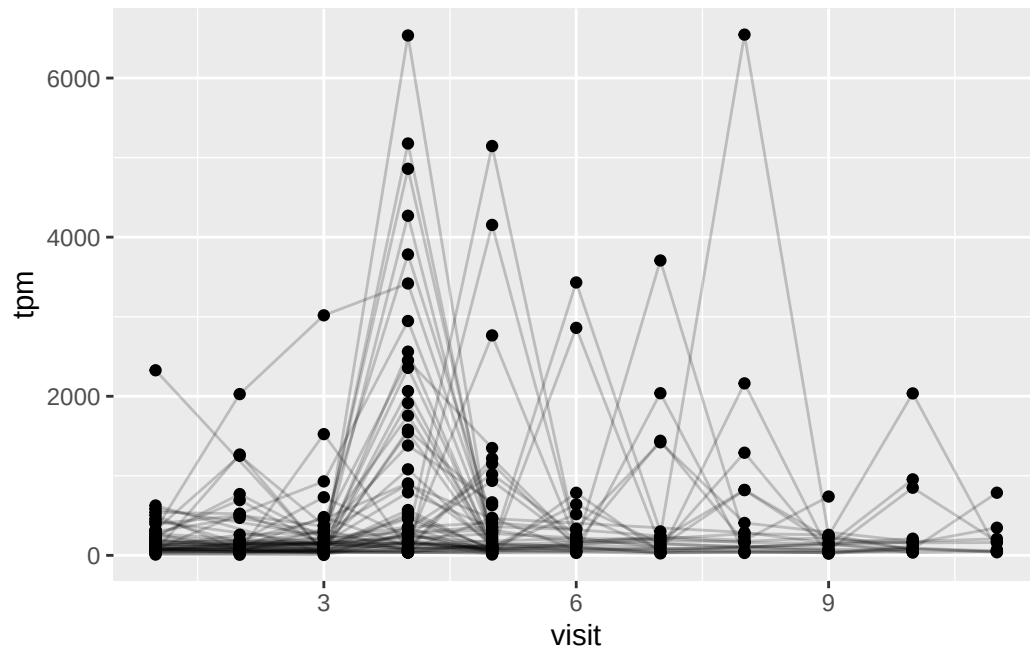
Obtaining CMI-PB RNASeq data

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)
```

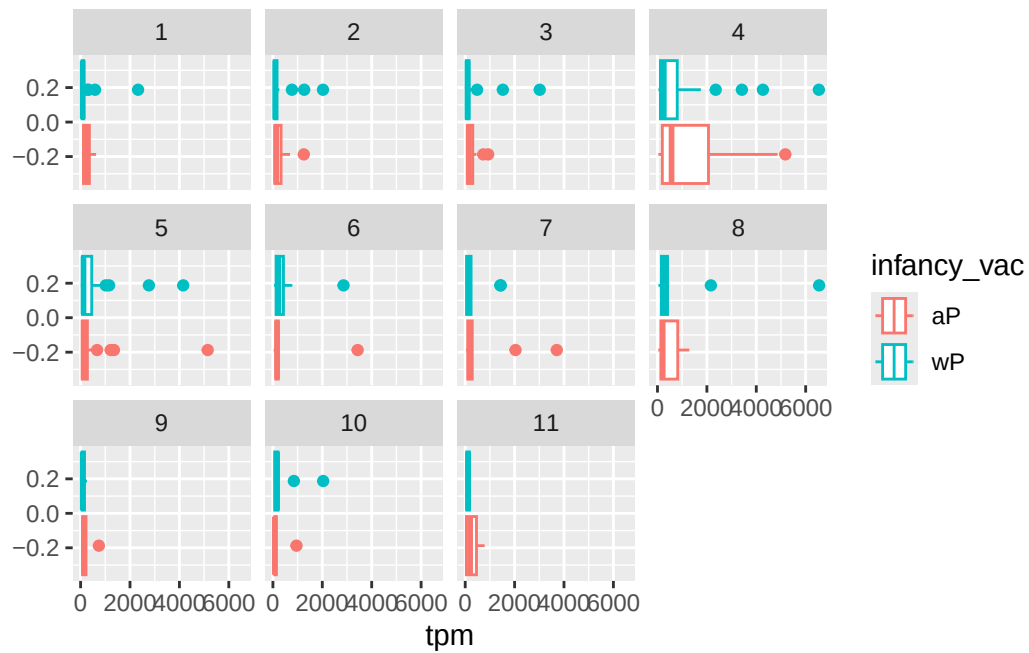
```
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
ggplot(ssrna) +
  aes(x = visit, y = tpm, group = subject_id) +
  geom_point() +
  geom_line(alpha = 0.2)
```



```
ggplot(ssrna) +  
  aes(tpm, col=infancy_vac) +  
  geom_boxplot() +  
  facet_wrap(vars(visit))
```



```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
    aes(tpm, col=infancy_vac) + geom_density() +
    geom_rug()
```

